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Artificial intelligence applications in aviation accident classification: A preliminary exploratory study

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ABSTRACT

Artificial intelligence (AI) applications are already integrated and expected to expand within the aviation sociotechnical ecosystem. Machine Learning (ML) models are capable and could excel on accident report classification tasks within the aviation context, assuming adequate and proper training. Human subject matter experts (SMEs) and the open access AI ChatGPT reanalyzed the TransAsia Airways flight GE235, Air France AF447, and Helios Airways Flight HCY552 accidents using the Human Factors Analysis and Classification System (HFACS) to investigate whether the AI agent is capable of conducting aviation accident analysis and comparing raters' results, of verifying AI's learning abilities, and, finally, investigating whether AI analysis could expand in other systems-based models. The results suggested that: (a) AI/ML applications could be an efficient tool for learning organizations' continuous learning, growth, and improvement by creating new knowledge from previous failures, (b) the proper prompts and well-established frameworks could optimize the agreement between human SMEs and AI in accident analysis, (c) the ChatGPT performed better on linear models (i.e., Bowtie) compared to complex systemic accident causation models (e.g., HFACS), (d) the ChatGPT could not conduct high- and low-level analysis simultaneously, and, (e) a multidisciplinary team including human factors, IT, and accident investigators experts is essential in developing and overseeing the W-shaped learning assurance process.

1. Introduction

Most accidents or incidents are not caused by a single factor but rather by a combination of interacting contributing factors [1]. Human Factors Analysis and Classification System (HFACS) is a systems-based accident framework utilized in the analysis and understanding of the human factors that contribute to accidents or incidents in complex systems, such as aviation, healthcare, and the military [2–4]. HFACS provides a way to systematically identify and categorize these contributing factors so that they can be analyzed and addressed [1,5]. Accident and incident analysis is promoted not only by regulations for threat identification [6–8] but also strengthens complex systems' continuous learning, growth, and improvement as a learning organization [9–11].

A learning organization can improve its ability to adapt to changing circumstances, innovate, and achieve its goals [10,12]. Organizations can improve their safety performance and reduce the risk of accidents or incidents by using appropriate methods and tools to create new knowledge from previous successes and failures [12,13]. Artificial intelligence (AI) applications are already integrated and expected to expand within the aviation sociotechnical ecosystem [14–16]. Machine learning models are capable and, assuming adequate and proper training, could excel on report classification tasks in Level 1 applications [17,18] within the aviation context [14].

Literature suggests that AI applications might have the potentials to contribute on organizational learning through adverse events analysis [19–21], and classification in a timely, efficient [22] and standardized fashion by increasing the validity while decreasing SME dependance [21,23]. This study reanalyzed the TransAsia Airways flight GE235, Air France AF447, and Helios Airways Flight HCY552 accidents using the human HFACS model by two different categories of raters: human subject matter experts (SMEs) and the open access AI ChatGPT [24]. The aim was to investigate whether the AI agent is capable of aviation accident analysis, to compare raters' results, to verify AI's learning abilities, and, finally, to investigate whether AI analysis could expand in other systems-based models.

2. Methodology

The authors acted as human factors analysts in the analysis and classification process for the TransAsia Airways flight GE235 [25], Air France AF447 [26], and Helios Airways Flight HCY552 accidents [27].

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Field	Criteria	AcciMap	Bow-Tie	EAST-BL	Fault Tree	FRAM	HFACS	STAMP-CAST	STEP
Systemic	Government Policy and Budgeting	X						X	
	Regulators and Associations	X		X				X	
	Local Area Government, Company Management	X		X	X	X	X	X	X
	Technical and Operational Management & Supervision	X	X	X	X	X	X	X	X
	Physical processes and actor activities	X	X	X	X	X	X	X	X
	Equipment and environment	X	X	X	X	X	X	X	X
Theoritical (7 Tenets)	Multiple contributory factors	X	X	X	X		X	X	X
	Multiple actors	X	X	X	X		X	X	X
	Interactions between factors	X		X	X	X			
	Vertical integration	X		X			X	X	
	Communications and feedback	X	X	X		X	X	X	X
	Migration of performancefrom safe to unsafe				X				X
	Triggering events	X	X		X	X	X	X	X
Methodological	Evidence of reliability	Medium	N/A	N/A	High	N/A	Medium	N/A	N/A
	Evidence of validity	Medium	N/A	N/A	High	N/A	Medium	N/A	N/A
	Complexity of approach	Low	Medium	Medium	Medium	High	Low	High	Low
	Reliannce on SME	Medium	Medium	High	High	Medium	Medium	High	Medium
	Auditability and traceability of system influences	High	Medium	Medium	Medium	Medium	Medium	Medium	Medium
	Applicability to analysis of accidents	High	Medium	Low	Medium	Low	Medium	High	High
	In-built classification scheme of contributory factors	No	No	No	No	No	Yes	Yes	No
Practical	Ease of use (high= easy)	High	Medium	Medium	Medium	Low	High	Low	High
	Application time (low=quick)	Low	Medium	Medium	Medium	High	Medium	High	Medium
	Training demand (low=little training)	Low	Medium	High	Medium	High	Low	Medium	Low
	Simplicity of interpretation (high=simple)	High	High	Medium	High	Low	High	Medium	High
	Tools required (low=pen and paper)	Low	Low	Low	Low	Medium	Medium	Medium	Low
	Evidence of practical impact (high=good)	Medium	High	Low	High	Medium	Medium	Medium	Medium

Fig. 1. Analysis method comparison based on systemic, theoretical, methodological, and practical criteria as adapted from Stanton et al. [23] with highlighted the method used for this study.
Notes: EAST–BL: Event Analysis of Systemic Teamwork – Broken Links; FTA: Fault Tree Analysis; FRAM: Functional Resonance Analysis Method; HFACS: Human Factors Analysis and Classification Scheme; STAMP–CAST: Systems–Theoretic Accident Model and Processes – Causal Analysis using Systems Theory; STEP: Sequential Timed Event Plotting.

Criteria for the study cases' accidents were: (a) controlled flight into terrain (CFIT) as causal factor, (b) final reports were released, (c) final reports were published in English language, (d) no previous published HFACS analysis was conducted and been available online on relevant academic databases (based on research and to authors' best knowledge to date of writing).

Stanton et al. [23] compared eight analysis methods based on systemic, theoretical, methodological, and practical criteria (Fig. 1). Specifically, the HFACS in this analysis made themes in causal factors easy to identify, enhanced reliability and could facilitate the further statistical process [28]. Moreover, it was expected that researchers' extensive experience and familiarity with the method would benefit reliability. However, HFACS constrained the classification to domain-specific types of failures [23] and predefined organizational levels [29], ruling out factors outside the involved organization and interactions between contributing factors. Thus, complimentary methods were utilized [5,30,31].

Based on the study's scope, the Person-Level and Mishap-Level Factors from HFACS version 7.0 were used, as adopted by the US Air Force (Air Force Safety Center, 2005) due to in-built classification scheme, ease of use, low-training demand (Fig. 1), and the researchers experience. In detail, Level-1 "Acts" included the "performance-based" or "judgement & decision-making errors" and "violations". Level-2 "Preconditions" included the subcategories "environment" (further divided into "physical" and "technological"), "physical and mental state" (further divided into "physical problems", "state of mind", "sensory misperception", and "mental awareness") and, lastly, "teamwork". Level-3 "Supervision" included the "Supervisory violations", "planned inappropriate operations", and "Inadequate supervision". Level-4, "Organizational influences", included the subcategories "resource problems", "personnel selection & staffing", "policy & process issues", and, lastly, "climate/culture influences".

Both analysts have been accident/incident investigators for the European Military Organization (EMO), each with ten (10) years of experience. They were trained in HFACS according to the EMO's regulations and the syllabus for the respective safety officers' courses. Analysts used the same data sources (i.e., official final reports) to support their unique form of analysis. The accidents were initially coded independently by each analyst following the framework's guidance [32]. Upon completing the initial analyses, the two analysts held Microsoft Teams' online meetings to exchange and review their outputs. Any discrepancies or disagreements were resolved through discussion until a consensus was reached.

Then, the research team prompted sequentially in iterative fashion the ChatGPT version February 2023 [24] to conduct an HFACS analysis for the respective case studies aviation accidents. For cohesion and to facilitate readers' comprehension the detailed prompts are included on the relevant results paragraphs (Sections 3.1.2, 3.2.2, and 3.3.2).

Moreover, ChatGPT prompted to conduct a Bowtie [33], an Accident Modeling- AcciMap [34], and a System- Theoretic Accident Modeling and Processes (STAMP) analysis [35] for TransAsia [13]. *Bowtie analysis* [33] is a risk management methodology that helps visualize an event's cause and effect. It uses a diagram that resembles a bow tie, with the event at the center and the causes and consequences on either side. AcciMap analysis [34] visually represents the events and factors leading up to an accident. STAMP [35] is a structured process for identifying and analyzing aviation safety risks. It consists of four steps: Identify, Analyze, Mitigate, and Evaluate. Finally, AI was prompted to suggest an accident analysis framework and mitigation measures to avoid similar accidents in the future. Similarly, for cohesion and to facilitate readers' comprehension the detailed prompts are included on the relevant paragraphs in Section 4.

3. Case studies results

3.1. TransAsia Airways (TNA) GE235 HFACS analysis

On February 4th 2015, the domestic flight TransAsia Airways (TNA) Flight 235 from Taipei to Kinmen, Taiwan, operated by an ATR 72–600,

crashed into the Keelung River. On board were 58 people, 15 of whom survived.

3.1.1. SMEs' HFACS analysis

The SMEs' HFACS analysis is provided in the following paragraphs.

- **Organizational:** Regarding latent failures on the organization level, TNA did not have a clearly documented company policy (OP003) communicating the requirement to reject the takeoff if the automatic takeoff power control system did not arm. Moreover, the TNA crew resource management (CRM) program was implemented without sufficient oversight (OP006). Also, first officer (FO) recruitment and selection process was inadequate with marginal performance (OS001).
- **Supervision:** Regarding latent failures on the supervision level, TNA did not follow its procedures (SV001) when selecting and training pilots for upgrade. Moreover, TNA failed to provide proper training (SI003) since the differences training program may not have been sufficient to ensure that TNA flight crews were competent. Furthermore, the operator did not store, adequately maintain, and evaluate the differences in training records and subsequently failed to identify unsafe tendencies and institute remedial actions (SI007).
- **Preconditions:** Regarding latent failures on the preconditions level, the intermittent signal discontinuity between the auto feather unit (AFU) number 2 and the torque sensor dictates that the reliability of the automated system created an unsafe situation (PE205). Display logic of flight directors could be included in the category as well. From the interviews, it could be inferred that the pilot flying (PF) might tend to rush and inadequately communicate procedures (PC205). Moreover, both operating crew members failed to obtain relevant data from the instrument readings (PC505) regarding the loss of power from both engines. The crew did not respond timely to stall warnings; thus, it could be supported that they should have been task saturated (PC103). Furthermore, the PF did not retain procedural knowledge after training (PC109), including his continued difficulties in handling emergency and/or abnormal situations, engine flameout at takeoff and single-engine operations. The PF did not appropriately respond to or integrate input from the pilot monitoring (PM) indicates both failure of crew team leadership (PP101) by PF and lack of assertiveness (PP105) on behalf of the PM. Additionally, ineffective communication (PP108) is detected.
- **Acts:** Regarding the active failures, the PF applied inappropriate flight control inputs (AE104), resulting in a series of stalls following the loss of power during the initial climb. Moreover, the crew did not apply the documented failure identification procedure (AE103) prior to executing any actions, and they did not comply with TNA ATR72-600 standard operating procedures associated with engine flameout at takeoff. Furthermore, the crew did not reject the takeoff, and the PF disconnected the autopilot; those could be regarded as the wrong choice of actions (AE206). The crew also failed to prioritize tasks adequately (AE202) by first stabilizing the aircraft flight path, properly identifying the malfunction, and then taking actions per the engine number 2 flameout at takeoff following the golden rules [36]. It could be argued that the crew ignored the stall warnings (AE205), too. Lastly, the crew worked around (AV001) the required pre-takeoff briefing resulting in them not being mentally prepared for the one engine inoperative procedures.

3.1.2. Chat GPT's HFACS analysis

In response to the user's prompt "*conduct an HFACS version 7.0 analysis for TransAsia airways flight GE235 based on the report released on February 2015*", the ChatGPT generated the following response:
"*Here is an HFACS Version 7.0 analysis of TransAsia Airways Flight GE235 based on the report released in February 2015:*

- ***Organizational Influences:***
  - *Inadequate safety culture within TransAsia Airways that did not prioritize safety, leading to a lack of adherence to standard procedures and guidelines.*
  - *Inadequate oversight and enforcement of safety regulations by the Civil Aeronautics Administration of Taiwan.*
  - *Poor resource management by TransAsia Airways, including inadequate training, insufficient maintenance, and suboptimal crew pairing.*
- ***Preconditions for Unsafe Acts:***
  - *Inadequate crew resource management, including ineffective communication, leadership, and decision-making among the flight crew.*
  - *Fatigue and inadequate rest due to scheduling and work-hour policies.*
  - *Insufficient training, experience, and proficiency of the flight crew.*
  - *Use of automation as a substitute for manual flying skills, leading to automation bias and skill degradation.*
- ***Unsafe Acts:***
  - *The flight crew's failure to recognize and respond appropriately to the aircraft's stall warning system, indicating a loss of situational awareness and ineffective monitoring of critical flight parameters.*
  - *Inappropriate use of engine thrust, resulting in an incorrect response to the engine malfunction.*
  - *Deviation from standard operating procedures, including failure to execute the proper landing checklist and procedures.*
  - *Poor decision-making, including the decision to continue the flight despite known mechanical issues and adverse weather conditions.*
- ***Outcomes:***
  - *Loss of the aircraft and all passengers and crew on board*
  - *Property damage and environmental impact from the crash*

Overall, the HFACS analysis of TransAsia Airways Flight GE235 based on the February 2015 report identifies multiple organizational, crew-related, and environmental factors that contributed to the accident. Addressing these factors through improved training, communication, and safety management systems could help prevent similar accidents in the future." [24].

3.2. Air France AF447 HFACS analysis

On June 1st, 2009, Air France flight 447 departed from Rio de Janeiro, Brazil, to Paris, France, with an Airbus A330 and crashed into the Atlantic Ocean.

3.2.1. SMEs' HFACS analysis

The SMEs' Air France AF447 HFACS analysis is provided in the following paragraphs.

- **Organizational:** European Aviation Safety Agency (EASA) had analyzed past events but decided not to make the probe change mandatory (OR008 & OP002) and indicates over-confidence in equipment (OC003). The copilots lacked any formal in-flight training on manual aircraft handling at high altitude and there was no regulatory training for a crew made up of two copilots with a relief captain (OP004).
- **Supervision:** The operator failed to provide proper training on high altitude aircraft handling (SI003). The two copilots teaming could be attributed to supervision level fallacy (SP001 & SP003). The stall training provided inaccurate expectations (PC110). A

few unwritten procedures had emerged and become established among pilots regarding the use of automation (SV002). Inadequate weather-wise risk assessment (SP006).

- **Preconditions:** Captain left the cockpit without communicating instructions for crossing the intertropical convergence zone (ICTZ); thus, inadequately briefed the two FOs (PP101) and showed lack of leadership (PP109). The measured speed difference denotes instrumentations' issues caused by the icing on pitot probes followed by no failure Electronic Centralized Aircraft Monitor (ECAM) messages (PE202). Autopilot disconnection created confusion to the pilots (PC104). Absence of visual cues contributed, too (PE101). Pilots awake time might be a contributing factor (PC307).
- **Acts:** The main causal factor was the inappropriate control inputs after autopilot disconnection (AE103) and wrong choice of actions (AE206). Neither of the pilots acknowledged the stall situation (AE205).

3.2.2. Chat GPT's HFACS analysis

In response to the user's prompt "*Here is the TNA HFCAS analysis by SMEs as feedback (text intentionally omitted; see Section 3.1.1 for details). Use it to conduct an HFACS version 7.0 analysis for Air France Flight AF447 based on the report released on February 2012 by BEA*", the ChatGPT generated the following response:

- ***"Organizational Influences:***
  - *Inadequate training and experience of the crew (OP006).*
  - *Ineffective communication between the crew and maintenance personnel (OP002).*
  - *Inadequate procedures for handling a malfunctioning airspeed indicator (OP008).*
  - *Inadequate feedback mechanisms to report and learn from safety-related incidents (OP005).*
- ***Preconditions:***
  - *Lack of experience in flying in turbulence and icing conditions (PC202).*
  - *Inadequate understanding of the aircraft systems and automation (PC204).*
  - *Inadequate understanding of the effects of high-altitude stall conditions (PC201).*
  - *Failure to effectively monitor and manage aircraft speed and altitude (PC103).*
- ***Supervision:***
  - *Inadequate supervision and oversight of crew training and performance (SV004).*
  - *Inadequate supervision of the implementation of safety programs and procedures (SV003).*
  - *Inadequate supervision of the maintenance of the aircraft and its systems (SV002).*
  - *Inadequate oversight of the Airbus A330 certification and the adequacy of its stall recovery procedures (SV001).*
- ***Unsafe Acts:***
  - *Failure to effectively monitor and manage aircraft speed and altitude, leading to a stall (AE201).*
  - *Inadequate crew resource management (CRM) and communication during the flight (AE202).*
  - *Misinterpretation of flight data by the crew (AE203).*
  - *Failure to follow the appropriate stall recovery procedures (AE204).*
  - *Inadequate crew response to stall warnings (AE205).*
  - *Inadequate crew response to airspeed indications (AE206).*
  - *Inadequate use of automation and autopilot (AE207).*

In conclusion, the HFACS analysis of the AF447 accident highlights several organizational, precondition, supervisory, and active failures that contributed to the accident. To prevent similar accidents in the future, measures could include enhancing crew training and experience, improving communication and CRM, enhancing oversight and supervision of safety programs and procedures, and improving the design and certification of aircraft systems and procedures for handling high-altitude stall conditions". [24].

In response to the user's prompt "*Conduct again an HFACS analysis for Air France Flight AF447 based on the report released on February 2012 by BEA with the following codes and their respective descriptions for each code as per HFACS version 7.0 and assign extracts from the report to the outcome codes:*

Code Description
AE101 Unintended Operation of Equipment
AE102 Checklist Not Followed Correctly
AE103 Procedure Not Followed Correctly
. . . (text intentionally omitted; for details refer to HFACS v.7.0)",
the ChatGPT generated the following response:

- *"PE201: Seat and Restraint System Problems - The accident occurred during a night flight over the Atlantic Ocean. The flight encountered turbulence and the aircraft's speed sensors became obstructed by ice crystals, causing the autopilot to disengage. This resulted in the aircraft entering a high-altitude stall, from which the pilots were unable to recover. The report noted that the seats and restraint systems on the Airbus A330 were not designed to withstand the forces of an inflight breakup, which likely contributed to the high number of fatalities.*
- *AE103: Procedure Not Followed Correctly - The report also found that the pilots did not follow proper procedures in response to the stall warning. They pulled the nose of the aircraft up, rather than down, which exacerbated the stall. Additionally, they did not communicate effectively with each other or use their instruments properly to diagnose the problem.*
- *AE105: Ignored a Caution/Warning - The flight crew received multiple warnings and alerts leading up to the stall, including airspeed discrepancies, but they did not respond appropriately.*
- *PC307: Fatigue - The report also identified fatigue as a contributing factor. The pilots had been awake for more than 24 h leading up to the flight, which may have impaired their judgment and performance.*
- *PP101: Failure of Crew/Team Leadership - The report found that there was a lack of clear leadership and communication within the cockpit, with the captain leaving the cockpit to take a nap shortly before the incident. This may have contributed to the breakdown in procedures and decision-making.*
- *SV001: Failure to Enforce Existing Rules (supervisory act of omission) - "This illustrates a failure to comply with the basic principle of supervisory responsibility which is to monitor the proper application of procedures and to take appropriate action if necessary." (Section 2.2.2)*
- *SV002: Allowing Unwritten Policies to Become Standard - "In view of the difficulties involved in the management of the automation system, a number of unwritten procedures had emerged and become established among pilots." (Section 1.17.2.2)*
- *SP001: Directed Task Beyond Personnel Capabilities - "The monitoring of the speed and altitude of the aircraft, the basic tasks of an aircraft commander, had been delegated to the co-pilot who, in this instance, lacked the necessary experience." (Section 1.16.1)*
- *SP003: Selected Individual with Lack of Current or Limited Experience - "The First Officer had 2,936 flying hours and had held a co-pilot position on an A330 for nearly one year, but he had little experience of flying in manual control at high altitude." (Section 1.16.2)*
- *SP006: Performed Inadequate Risk Assessment – Formal - "Air France's procedures for risk assessment prior to the departure of a flight were not adequate. They did not identify that the weather conditions in the area to be crossed could be particularly dangerous." (Section 2.2.2)*

- *SI001: Supervisory/Command Oversight Inadequate - "The captain did not retake control of the aircraft, despite the copilot's contradictory and incorrect actions, nor did he adequately inform the copilot of the situation, thus contributing to the continuation of the inappropriate actions." (Section 1.16.2)*
- *SI003: Failed to Provide Proper Training - "Air France's training program for manual flying at high altitude is not adapted to the Airbus A330/340 and is not sufficient to enable pilots to acquire the necessary reflexes in the event of an unexpected event, especially one which has never been experienced before." (Section 2.2.2)*
- *OP002: Organizational Program/Policy Risks not Adequately Assessed - "The approach to crew training developed by Air France for the A330/340 family does not fully take into account the specific features of this aircraft in terms of the interaction between systems, or the handling of the aircraft in alternate law mode." (Section 2.2.2)*
- *OC001: Organizational Culture (attitude/actions) Allows for Unsafe Task/Mission - "The prevailing culture at Air France at the time of the accident was one of considerable deference to authority. This may have had an impact on the ability of the crew to manage the unexpected situation they encountered." (Section 2.2.2)*
- *OC002: Organizational Over-confidence or Under-confidence in Equipment - "The crew did not appear to have an understanding of the limitations of the pitot probes, or the risks associated with high-altitude flight without reliable airspeed information". (Section 1.15.3.3)"."* [24].

3.3. Helios Airways Flight HCY552 HFACS analysis

Lastly, the Helios Airways Flight HCY552 accident occurred on August 14, 2005, when a Boeing 737–300 crashed into a mountain near Grammatiko, Greece, killing all 121 passengers and crew members.

3.3.1. SMEs' HFACS analysis

The SMEs' analysis is provided in the following paragraphs.

- **Organizational:** The airline's management did not provide adequate oversight (SI001). Moreover, the airline's command and control resources were deficient (OR001) and Cyprus Department of Civil Aviation (DCA) lacked qualified personnel (OS002). The operator has not provided hypoxia training (SI003). Also, the organizational culture within the airline was conducive to non-compliance with established safety procedures and protocols (OC001).
- **Supervision:** There was inadequate risk assessment (SP006).
- **Preconditions:** The First Officer did not retain procedural knowledge after training (PC109). The captain failed to ensure that the pre-flight checklist was completed correctly (PP101). The crew became confused, and task saturated with the equipment cooling fan situation (PC103 & PC104). The use of the same aural warning for two different trigger events indicates warning system issues (PE202). The crew misinterpreted the auditory cues (PC507). There was lack of effective communication between the crew members in the malfunction diagnosis (PP108). The lack of pressurization led to hypoxia (PC312) and eventually loss of consciousness (PC304).
- **Acts:** The flight crew did not complete the pre-flight checklist (AE102), ignored the warning indications (AE105), and did not follow the correct procedures when responding to the pressurization system failure (AE103). The crew failed to recognize the seriousness of the situation (AE201) and followed wrong choice of action (AE206).

3.3.2. Chat GPT's HFACS analysis

In response to the user's prompt "*Please conduct again an HFACS analysis for Helios Airways Flight HCY522 based on the report released on November 2006 by AAIASB with the following codes and their respective descriptions for each code as per HFACS version 7.0 and assign extracts from the report to the outcome codes:*

Code Description
AE101 Unintended Operation of Equipment
AE102 Checklist Not Followed Correctly
AE103 Procedure Not Followed Correctly
. . . (text intentionally omitted; for details refer to HFACS v.7.0)",

the ChatGPT generated the following response:

- *"AE102: Checklist Not Followed Correctly - The flight crew did not complete the pre-flight checklist and failed to recognize that the pressurization system was set to manual instead of automatic mode. This error led to the aircraft cabin not pressurizing properly, which caused the crew and passengers to experience hypoxia and eventually led to the loss of consciousness of the crew.*
- *AE103: Procedure Not Followed Correctly - The flight crew did not follow the correct procedures when responding to the pressurization system failure. They misinterpreted the warning indications, and as a result, they were unable to take the necessary corrective actions.*
- *AE105: Ignored a Caution/Warning - The flight crew ignored the warning indications and failed to respond to the aircraft's systems and procedures indicating that the pressurization system was not functioning correctly.*
- *AE201: Inadequate Real-Time Risk Assessment - The flight crew failed to perform an adequate real-time risk assessment, which would have identified the pressurization system failure and the resulting cabin depressurization. The crew failed to recognize the seriousness of the situation, and the delay in taking appropriate action led to the loss of the aircraft and all onboard.*
- *AV001: Performs Work-Around Violation - The flight crew attempted to perform a work-around by manually controlling the pressurization system, which was not the correct procedure. This work-around led to the aircraft cabin not pressurizing properly, which caused the crew and passengers to experience hypoxia.*
- *PC307: Fatigue - The captain was fatigued at the time of the accident, which likely contributed to his failure to respond appropriately to the situation.*
- *PC508: Spatial Disorientation - The flight crew likely experienced spatial disorientation due to the loss of consciousness resulting from hypoxia, which may have prevented them from recognizing the aircraft's attitude and altitude.*
- *PP101: Failure of Crew/Team Leadership - The captain failed to ensure that the pre-flight checklist was completed correctly, and he did not recognize the pressurization system's failure or take appropriate corrective action. The captain's failure to lead his crew contributed to the accident.*
- *SI001: Supervisory/Command Oversight Inadequate - The airline's management did not provide adequate oversight or support to ensure that the flight crew followed proper procedures and protocols. The airline did not have an effective safety management system in place to prevent such accidents.*
- *OR001: Command and Control Resources are Deficient - The airline's command and control resources were deficient, as the airline did not have an effective safety management system in place to identify and mitigate safety risks.*
- *OC001: Organizational Culture (attitude/actions) Allows for Unsafe Task/Mission - The organizational culture within the airline was conducive to non-compliance with established safety procedures and protocols. The airline did not place a high enough value on safety and did not provide adequate resources or training to ensure that safety protocols were followed.*

In conclusion, the HFACS analysis of the Helios Airways Flight HCY552 accident indicates that multiple factors contributed to the accident, including the flight crew's failure to follow proper procedures and protocols, inadequate real-time risk assessment, fatigue, spatial disorientation, ineffective

crew/team leadership, inadequate oversight and support from management, deficient command and control resources, and an organizational culture that allowed for non-compliance with established safety procedures and protocols." [24].

4. Results on extended functions

4.1. AI's generated Bowtie analysis

In response to the user's prompt "*conduct a Bowtie analysis for TransAsia airways flight GE235 based on the report released on February 2015*", the ChatGPT generated the following response:

"Here is a bow tie analysis for the TransAsia Airways Flight GE235 accident:

Event: *TransAsia Airways Flight GE235 crashed into the Keelung River during takeoff on February 4, 2015.*

Threats/Hazards:

- *Mechanical failure in engine #2.*
- *Inadequate training and experience of the pilots.*
- *Inadequate communication between the pilots.*
- *Lack of adherence to standard operating procedures (SOPs).*
- *Lack of effective supervision and oversight by TransAsia Airways.*

Prevention Barriers:

- *Maintenance procedures for engine #2.*
- *Effective training and experience of the pilots.*
- *Clear and effective communication protocols between the pilots.*
- *Adherence to SOPs.*
- *Effective supervision and oversight by TransAsia Airways.*

Mitigation Barriers:

- *Emergency procedures to handle engine failure during takeoff.*
- *Backup systems and redundancy to prevent loss of control.*
- *Effective CRM to improve communication and coordination between the pilots.*
- *Proper and timely decision-making by the pilots.*
- *Effective oversight and regulatory enforcement by aviation authorities.*

Consequences:

- *Loss of life and injuries to passengers and crew.*
- *Damage to the aircraft and the environment.*
- *Reputational damage to TransAsia Airways and the aviation industry.*
- *Financial losses for the airline, passengers, and stakeholders.*
- *Legal and regulatory consequences, including investigations and enforcement actions.*

By analyzing the threats/hazards, prevention barriers, mitigation barriers, and consequences, this bow tie analysis can help identify the critical areas where improvements can be made to prevent similar accidents in the future. The identified prevention and mitigation barriers can be strengthened to reduce the likelihood and severity of the threats/hazards, and the consequences can be minimized by effective emergency response and management." [24].

4.2. AI's generated AcciMap analysis

In response to the user's prompt "*conduct an AcciMap analysis for TransAsia airways flight GE235 based on the report released on February 2015*", the ChatGPT generated the following response:

"Here's an AcciMap analysis for the TransAsia Airways Flight GE235 accident:

Direct causes:

- *Inappropriate flight control inputs by the pilot flying (PF).*
- *Failure to apply documented failure identification procedure.*
- *Non-compliance with standard operating procedures associated with engine flameout at takeoff.*
- *Failure to reject takeoff.*
- *Disconnection of autopilot.*
- *Failure to prioritize tasks adequately.*
- *Ignoring stall warnings.*
- *Workaround of required pre-takeoff briefing.*

Contributing factors:

- *Intermittent signal discontinuity between the auto feather unit (AFU) number 2 and the torque sensor.*
- *Display logic of flight directors.*
- *Rushing and inadequate communication of procedures by the PF.*
- *Failure to obtain relevant data from instrument readings.*
- *Failure to respond timely to stall warnings.*
- *Inadequate retention of procedural knowledge after training.*
- *Failure of crew team leadership by PF.*
- *Lack of assertiveness on behalf of the PM.*
- *Ineffective communication.*
- *Insufficient differences training program to ensure competency.*
- *Failure to store, adequately maintain, and evaluate differences in training records.*
- *Non-compliance with company policy communicating the requirement to reject takeoff if the automatic takeoff power control system did not arm.*
- *Insufficient oversight of the CRM program.*

Underlying causes:

- *Insufficient pilot training and qualification procedures.*
- *Inadequate maintenance and inspection of the aircraft.*
- *Insufficient oversight of the operator's procedures and practices by the regulatory agency.*
- *Inadequate risk assessment and management by the operator.*

Consequences:

- *Loss of the aircraft.*
- *43 fatalities.*
- *Environmental damage.*
- *Financial losses for the operator and other parties involved."* [24].

In response to the user's prompt "*conduct an AcciMap analysis for TransAsia airways flight GE235 based on the report released on February 2015 using the following framework... (intentionally omitted)*", the ChatGPT generated the following response:

"***Government policy and budgeting:***

- *Funding and resources allocated to aviation safety programs.*
- *Regulatory requirements for airlines and aircraft maintenance.*
- *Standards for pilot training and qualification.*
- *Air traffic control procedures and technology.*
- *Policies and procedures for emergency response and search and rescue operations.*

Regulatory bodies and associations:

- *The role of the Civil Aeronautics Administration (CAA) in oversight of the aviation industry in Taiwan.*
- *Compliance and enforcement actions taken by the CAA in response to safety violations.*
- *The effectiveness of safety management systems (SMS) in identifying and mitigating safety risks.*
- *International regulatory standards and best practices.*

Local health economy planning and budgeting:

- *Availability and accessibility of medical services in the area surrounding the accident site.*

```
                    Government policy and budgeting
                                  |
             ----------------------------------------
            |                |                      |
Regulatory bodies   Local health economy planning     Technical and
and associations          and budgeting            operational management
            |                |                      |
      Human factors     Hospital readiness        Pilot training
            |                                       |
      Fatigue and                                     |
    workload issues                              Engine failure
            |                                       |
    Weather conditions                             |
            |                                       |
    Aircraft maintenance                           |
            |                                       |
       Communication                             Events, processes
            |                                 and conditions
    Runway overruns,                               |
    undershoots, and                            Spatial disorientation
    other incidents                                |
            |                                       |
         Outcomes                             Pilot error (decision
                                                 making)
```

Fig. 2. AcciMap diagram as produced by ChatGPT [24].

- *Adequacy of emergency response resources and training.*
- *Coordination and communication between medical personnel and emergency responders.*

Technical and operational management:

- *Maintenance records and practices for the aircraft involved in the accident.*
- *Adequacy of training and proficiency of the flight crew, including captain and first officer.*
- *Airline policies and procedures related to engine failure and emergency response.*
- *Operational pressures and demands on the flight crew, such as scheduling and work hours.*

Events, processes, and conditions:

- *Weather conditions at the time of the accident.*
- *Communication and coordination between the flight crew and air traffic control.*
- *The decision-making process of the flight crew during the emergency.*
- *The physical and psychological state of the flight crew during the emergency.*

Outcomes:

- *The number and severity of injuries and fatalities.*
- *The response of emergency services and search and rescue teams.*
- *The impact of the accident on the airline industry in Taiwan.*
- *The effectiveness of the investigation and identification of contributing factors to prevent future accidents.”* [24].

In response to the user’s prompt “*produce an AcciMap diagram for TransAsia airways flight GE235 based on AcciMap analysis*”, the ChatGPT generated the following response as seen in Fig. 2.

4.3. AI’s generated STAMP analysis

In response to the user’s prompt “*conduct a STAMP analysis for TransAsia airways flight GE235 based on the report released on February 2015*”, the ChatGPT generated the following response:
“Here is a STAMP analysis:

Identify the top event:

The top event is the crash of TransAsia Airways Flight GE235 during takeoff on February 4, 2015.

Identify the hazards:

- *Inadequate training and experience of the flight crew.*
- *Inadequate procedures for handling engine failure.*
- *Inadequate maintenance of the aircraft.*
- *Lack of automation in the cockpit.*
- *Inadequate oversight by the regulatory authorities.*
- *Inadequate communication between the flight crew.*

Identify the system components:

- *Flight crew.*
- *Aircraft.*
- *Maintenance procedures.*
- *Regulatory authorities.*
- *Communication procedures.*
- *Training and experience requirements.*

Identify the control structure:

- *Flight crew training and experience requirements.*
- *Maintenance procedures and oversight.*
- *Regulatory oversight and enforcement.*
- *Communication procedures and protocols.*

Identify the control actions:

- *Improved training and experience requirements for flight crews.*
- *Improved procedures for handling engine failure.*

- *Improved maintenance procedures and oversight.*
- *Increased automation in the cockpit.*
- *Improved regulatory oversight and enforcement.*
- *Improved communication procedures and protocols.*

Analyze the interactions and dependencies:

- *Inadequate training and experience of the flight crew contributed to their inability to handle the engine failure.*
- *Inadequate procedures for handling engine failure contributed to the confusion and lack of coordination in the cockpit.*
- *Inadequate maintenance of the aircraft contributed to the engine failure.*
- *Lack of automation in the cockpit contributed to the flight crew's inability to quickly assess and respond to the situation.*
- *Inadequate oversight by the regulatory authorities contributed to the lack of enforcement of safety procedures.*
- *Inadequate communication between the flight crew contributed to the confusion and lack of coordination in the cockpit.*

Identify the system-level factors:

- *Inadequate training and experience requirements for flight crews.*
- *Inadequate procedures for handling engine failure.*
- *Inadequate maintenance procedures and oversight.*
- *Lack of automation in the cockpit.*
- *Inadequate regulatory oversight and enforcement.*
- *Inadequate communication procedures and protocols.*

Develop recommendations:

- *Increase the training and experience requirements for flight crews.*
- *Improve the procedures for handling engine failure.*
- *Strengthen maintenance procedures and oversight.*
- *Increase the use of automation in the cockpit.*
- *Strengthen regulatory oversight and enforcement.*
- *Improve communication procedures and protocols between the flight crew and ground personnel.*

STAMP analysis can be visualized as a hierarchical diagram, which shows the relationships and dependencies between the system components, control structure, control actions, and system-level factors." [24].

In response to the user's prompt "*produce a STAMP hierarchical diagram including report extracts based on your STAMP analysis*", the ChatGPT generated the following response as seen in Fig. 3.

System Level

Inadequate Safety Culture
(Chapter 2.2, p. 17-18)

STAMP Model

Work Environment Factors

Lack of Adequate Training (Chapter 2.4, p. 19-20)
Insufficient Maintenance of Aircraft (Chapter 2.7, p. 23-25)

Control Factors

Inadequate Pilot Monitoring (Chapter 2.5, p. 20-22)
Loss of Situational Awareness (Chapter 2.6, p. 22-23)

Process Factors

Ineffective Standard Operating Procedures (Chapter 2.5, p. 21)
Lack of Standardization (Chapter 2.5, p. 21)

Event Factors

Engine Flameout (Chapter 2.3, p. 18)
Instrument Malfunction (Chapter 2.3, p. 18-19)

Outcome

Aircraft Crash
(Chapter 1, p. 4-5)

Fig. 3. STAMP hierarchical diagram including report extracts as produced by ChatGPT [24].

4.4. Suggested analysis framework

In response to the user's prompt "*based on your accident analyses so far, can you suggest an accident analysis and classification framework*", the ChatGPT generated the following response:

"*Sure, here's a framework for analyzing aviation accidents:*

1. ***Contextual factors:***
   - *Government policies and budgeting.*
   - *Regulatory bodies and associations.*
   - *Local health economy planning and budgeting.*
   - *Technical and operational management.*
2. ***Human factors:***
   - *Crew resource management.*
   - *Training and experience of crew and maintenance personnel.*
   - *Human error and decision-making.*
   - *Workload and stress.*
3. ***Technical factors:***
   - *Aircraft design and maintenance.*
   - *Avionics and systems.*
   - *Weather conditions and forecasts.*
   - *Air traffic control and communications.*
4. ***Operational factors:***
   - *Standard operating procedures.*
   - *Emergency response and contingency planning.*
   - *Airline and airport culture and safety climate.*
   - *Organizational factors such as management and leadership.*
5. ***Outcome factors:***
   - *Immediate and long-term effects on passengers, crew, and other parties.*
   - *Financial impact on airlines and other stakeholders.*
   - *Public perception and media coverage.*
   - *Lessons learned and improvements made to prevent future accidents.*

This framework considers both the immediate and underlying causes of aviation accidents, including factors related to the broader aviation industry, human factors, technical factors, operational factors, and outcome factors. By examining accidents through each of these lenses, investigators can gain a more comprehensive understanding of the factors contributing to the accident and identify opportunities for improvement" [24].

4.5. Safety recommendations

In response to the user's prompt regarding TNA "*based on your analysis which mitigation measures can you suggest avoiding similar accidents in the future?*", the ChatGPT generated the following response:

"Based on the HFACS analysis of TransAsia Airways Flight GE235, the following mitigation measures could be implemented to help prevent similar accidents in the future:

Table 1
Comparative tables between SMEs' and AI's HFACS analysis for TransAsia Airways flight GE235.

TransAsia Airways Flight GE235	
Human SMEs	ChatGPT
Organizational influences	
• TNA did not have a clearly documented company policy (**OP003**) communicating the requirement to reject the takeoff. • TNA CRM program was implemented without sufficient oversight (**OP006**). • FO recruitment and selection process was inadequate with marginal performance (**OS001**).	• Inadequate safety culture within TransAsia Airways that did not prioritize safety, leading to a lack of adherence to standard procedures and guidelines. • Inadequate oversight and enforcement of safety regulations by the Civil Aeronautics Administration of Taiwan. • Poor resource management by TransAsia Airways, including inadequate training, insufficient maintenance, and suboptimal crew pairing.
Supervision	
• TNA did not follow its procedures (**SV001**) when selecting and training pilots for upgrade. • TNA failed to provide proper training (**SI003**) since the differences training program may not have been sufficient to ensure that TNA flight crews were competent. • The operator did not store, adequately maintain, and evaluate the differences in training records and subsequently failed to identify unsafe tendencies and institute remedial actions (**SI007**).	
Preconditions	
• The reliability of the automated system created an unsafe situation (**PE205**). Display logic of flight directors could be included in the category as well. • The PF might tend to rush and inadequately communicate procedures (**PC205**). • Both operating crew members failed to obtain relevant data from the instrument readings (**PC505**) regarding the loss of power from both engines. • The crew did not respond timely to stall warnings; thus, it could be supported that they should have been task saturated (**PC103**). • The PF did not retain procedural knowledge after training (**PC109**), including his continued difficulties in handling emergency and/or abnormal situations, engine flameout at takeoff and single-engine operations. • The pilot flying did not appropriately respond to or integrate input from the pilot monitoring indicates both failure of crew team leadership (**PP101**) by PF and lack of assertiveness (**PP105**) on behalf of the PM. • Ineffective communication (**PP108**) is detected.	• Inadequate crew resource management, including ineffective communication, leadership, and decision-making among the flight crew. • Fatigue and inadequate rest due to scheduling and work-hour policies. • Insufficient training, experience, and proficiency of the flight crew. • Use of automation as a substitute for manual flying skills, leading to automation bias and skill degradation.
Acts	
• PF applied inappropriate flight control inputs (**AE104**), resulting in a series of stalls following the loss of power during the initial climb. • The crew did not apply the documented failure identification procedure (**AE103**) prior to executing any actions, and they did not comply with TNA ATR72-600 standard operating procedures associated with engine flameout at takeoff. • The crew did not reject the takeoff, and the PF disconnected the autopilot; those could be regarded as the wrong choice of actions (**AE206**). • The crew also failed to prioritize tasks adequately (**AE202**) by first stabilizing the aircraft flight path, properly identifying the malfunction, and then taking actions per the engine number 2 flameout at takeoff following the golden rules [36]. • The crew ignored the stall warnings (**AE205**). • The crew worked around (**AV001**) the required pre-takeoff briefing resulting in them not being mentally prepared for the one engine inoperative procedures.	• The flight crew's failure to recognize and respond appropriately to the aircraft's stall warning system, indicating a loss of situational awareness and ineffective monitoring of critical flight parameters. • Inappropriate use of engine thrust, resulting in an incorrect response to the engine malfunction. • Deviation from standard operating procedures, including failure to execute the proper landing checklist and procedures. • Poor decision-making, including the decision to continue the flight despite known mechanical issues and adverse weather conditions.

- *"Enhance pilot training programs: TNA should review and enhance its pilot training programs, particularly for upgrading pilots to new aircraft types. The differences training program should be designed to ensure that flight crews are competent in handling emergency and abnormal situations, engine flameout at takeoff, and single-engine operations.*
- *Improve crew resource management (CRM): TNA should implement a CRM program with sufficient oversight to ensure its effectiveness in preventing accidents caused by crew errors. The program should focus on improving crew communication, task prioritization, and assertiveness, particularly in emergency situations.*
- *Revise procedures: TNA should review and revise its standard operating procedures associated with engine flameout at takeoff, failure identification, and rejection of takeoff if the automatic takeoff power control system does not arm. The procedures should be clearly documented and communicated to all flight crews.*
- *Enhance maintenance and evaluation programs: TNA should improve its maintenance and evaluation programs to ensure the reliability and safety of its automated systems. The differences in training records should be adequately stored, maintained, and evaluated to identify unsafe tendencies and institute remedial actions.*
- *Ensure proper equipment design and display logic: The design and display logic of flight directors and auto feather units should be reviewed to ensure they are reliable and safe. Intermittent signal discontinuity between the auto feather unit and torque sensor should be detected and corrected to prevent unsafe situations.*

By implementing these mitigation measures, TNA can improve its safety performance and prevent similar accidents from occurring in the future". [24].

5. Discussion

The comparative tables (Tables 1, 2 & 3) between SMEs' and AI's analyses are provided to facilitate the discussion.

HFACS SMEs' preliminary analysis of TNA GE235, AF447, and HCY522 accidents reached a "near-perfect agreement" (Cohen's Kappa, $\kappa 1 = 0.83$, $\kappa 2 = 0.85$, and $\kappa 3 = 0.87$) which could be attributed to common SMEs background in training and experience [37]. Quantitative analysis between AI and human HFACS raters was performed only on the last iteration (Helios accident). From a qualitative perspective, the authors followed the three key dimensions of the generic guidelines on ChatGPT evaluation [38]. It could be noted (Table 1) that ChatGPT

Table 2
Comparative tables between SMEs' and AI's HFACS analysis for Air France flight AF447.

Air France Flight AF447		
Human SMEs	ChatGPT	ChatGPT after iterations
Organizational influences		
EASA had analyzed past events but decided not to make the probe change mandatory (**OR008** & **OP002**) and indicates over-confidence in equipment (**OC003**). The copilots lacked any formal in-flight training on manual aircraft handling at high altitude and there was no regulatory training for a crew made up of two copilots with a relief captain (**OP004**).	Inadequate training and experience of the crew (**OP006**). Ineffective communication between the crew and maintenance personnel (**OP002**). Inadequate procedures for handling a malfunctioning airspeed indicator (**OP008**). Inadequate feedback mechanisms to report and learn from safety-related incidents (**OP005**).	**OP002: Organizational Program/Policy Risks not Adequately Assessed** "The approach to crew training developed by Air France for the A330/340 family does not fully take into account the specific features of this aircraft in terms of the interaction between systems, or the handling of the aircraft in alternate law mode." (Section 2.2.2) **OC001**: **Organizational Culture (attitude/actions) Allows for Unsafe Task/Mission** "The prevailing culture at Air France at the time of the accident was one of considerable deference to authority. This may have had an impact on the ability of the crew to manage the unexpected situation they encountered." (Section 2.2.2) **OC002: Organizational Over-confidence or Under-confidence in Equipment** "The crew did not appear to have an understanding of the limitations of the pitot probes, or the risks associated with high-altitude flight without reliable airspeed information." (Section 1.15.3.3).
Supervision		
The operator failed to provide proper training on high altitude aircraft handling (**SI003**). The two copilots teaming could be attributed to supervision level fallacy (**SP001** & **SP003**). The stall training provided inaccurate expectations (**PC110**). A few unwritten procedures had emerged and become established among pilots regarding the use of automation (**SV002**). Inadequate weather-wise risk assessment (**SP006**).	Inadequate supervision and oversight of crew training and performance (**SV004**). Inadequate supervision of the implementation of safety programs and procedures (**SV003**). Inadequate supervision of the maintenance of the aircraft and its systems (**SV002**). Inadequate oversight of the Airbus A330 certification and the adequacy of its stall recovery procedures (**SV001**).	**SV001: Failure to Enforce Existing Rules (supervisory act of omission)** "This illustrates a failure to comply with the basic principle of supervisory responsibility which is to monitor the proper application of procedures and to take appropriate action if necessary." (Section 2.2.2) **SV002: Allowing Unwritten Policies to Become Standard** "In view of the difficulties involved in the management of the automation system, a number of unwritten procedures had emerged and become established among pilots." (Section 1.17.2.2) **SP001: Directed Task Beyond Personnel Capabilities** "The monitoring of the speed and altitude of the aircraft, the basic tasks of an aircraft commander, had been delegated to the co-pilot who, in this instance, lacked the necessary experience." (Section 1.16.1) **SP003: Selected Individual with Lack of Current or Limited Experience** "The First Officer had 2936 flying hours and had held a co-pilot position on an A330 for nearly one year, but he had little experience of flying in manual control at high altitude." (Section 1.16.2) **SP006: Performed Inadequate Risk Assessment – Formal** "Air France's procedures for risk assessment prior to the departure of a flight were not adequate. They did not identify that the weather conditions in the area to be crossed could be particularly dangerous." (Section 2.2.2) **SI001: Supervisory/Command Oversight Inadequate** "The captain did not retake control of the aircraft, despite the copilot's contradictory and incorrect actions, nor did he adequately inform the copilot of the situation, thus contributing to the continuation of the inappropriate actions." (Section 1.16.2) **SI003: Failed to Provide Proper Training** "Air France's training program for manual flying at high altitude is not adapted to the Airbus A330/340 and is not sufficient to enable pilots to acquire the necessary reflexes in the event of an unexpected event, especially one which has never been experienced before." (Section 2.2.2)

(*continued on next page*)

Table 2 (*continued*).

Preconditions		
Captain left the cockpit without leaving specific instructions for crossing the ICTZ; thus inadequately briefed the two first officers (PP101) and showed lack of leadership (**PP109**). The measured speed difference denotes instrumentations' issues caused by the icing on pitot probes followed by no failure ECAM messages (**PE202**). Autopilot disconnection created confusion to the pilots (**PC104**). Absence of visual cues contributed, too (**PE101**). Pilots awake time might be a contributing factor (**PC307**).	Lack of experience in flying in turbulence and icing conditions (**PC202**). Inadequate understanding of the aircraft systems and automation (**PC204**). Inadequate understanding of the effects of high-altitude stall conditions (**PC201**). Failure to effectively monitor and manage aircraft speed and altitude (**PC103**).	**PE201: Seat and Restraint System Problems** - The accident occurred during a night flight over the Atlantic Ocean. The flight encountered turbulence and the aircraft's speed sensors became obstructed by ice crystals, causing the autopilot to disengage. This resulted in the aircraft entering a high-altitude stall, from which the pilots were unable to recover. The report noted that the seats and restraint systems on the Airbus A330 were not designed to withstand the forces of an inflight breakup, which likely contributed to the high number of fatalities. **PC307: Fatigue** - The report also identified fatigue as a contributing factor. The pilots had been awake for more than 24 h leading up to the flight, which may have impaired their judgment and performance. **PP101: Failure of Crew/Team Leadership** - The report found that there was a lack of clear leadership and communication within the cockpit, with the captain leaving the cockpit to take a nap shortly before the incident. This may have contributed to the breakdown in procedures and decision-making.
Acts		
The main causal factor was the inappropriate control inputs after autopilot disconnection (**AE103**) and wrong choice of actions (**AE206**). Neither of the pilots acknowledged the stall situation (**AE205**).	Failure to effectively monitor and manage aircraft speed and altitude, leading to a stall (**AE201**). Inadequate crew resource management (CRM) and communication during the flight (**AE202**). Misinterpretation of flight data by the crew (**AE203**). Failure to follow the appropriate stall recovery procedures (**AE204**). Inadequate crew response to stall warnings (**AE205**). Inadequate crew response to airspeed indications (**AE206**). Inadequate use of automation and autopilot (**AE207**).	**AE103: Procedure Not Followed Correctly** - The report also found that the pilots did not follow proper procedures in response to the stall warning. They pulled the nose of the aircraft up, rather than down, which exacerbated the stall. Additionally, they did not communicate effectively with each other or use their instruments properly to diagnose the problem. **AE105: Ignored a Caution/Warning** - The flight crew received multiple warnings and alerts leading up to the stall, including airspeed discrepancies, but they did not respond appropriately.

identified the high levels but vaguely and not correlated with the specific HFACS subcodes in its first response. No contributing factors were identified on the supervisory level, and inconsistencies on the "acts" level. It might be supported that the accident analysis was generic with superficial consideration of the HFACS framework. Then, the first response for the AF4447 accident after the provided SMEs analysis results for TNA GE235 was again generic with irrationally assigned subcodes (Table 2). Based on a prompt in which explicitly provided framework codes were submitted, the second attempt resulted in acceptable analysis with properly assigned coding and supporting text extracts from the final accident report (Table 2). However, perhaps a more in-depth analysis would be desirable. In the last analysis (Helios HCY522 – Table 3), there was a fair agreement between human SMEs and AI (Cohen's Kappa, $\kappa = 0.37$), partly owed to more exhaustive human raters' analysis [37].

The prompt to a Bowtie analysis resulted in a detailed and fairly accurate response. However, the prompt for a more complex model like AcciMap resulted in a generic response with no relevance to requested framework. Upon providing feedback, the analysis followed the proper model, and it was satisfying, albeit with some inconsistencies—the produced diagram needed to be revised as expected from the respective application [38]. The STAMP analysis was process-oriented, resulting in logical findings, whilst the produced hierarchical diagram could be characterized as an acceptable attempt for data visualization considering the platform's limitations [38]. The suggested framework was relevant with some identified limitations; nonetheless it might be worth of further investigation. The mitigation measures were accurate and turned from generic to more specific upon iterative feedback.

On this preliminary study, the research team approached the ChatGPT utilization from a human factors perspective, they had no formal training in machine learning (ML) applications or prompt engineering [39–41] and followed an empirical non-systematic approach regarding their iterative feedback to ChatGPT. The development of AI/ML, accident analysis/classification model, should be based on trustworthiness gears to assure human factors compliance regarding the accuracy, reliability, explainability and unfair bias [14,42]. Moreover, supervised learning following the suggested W-shaped iterative process (Fig. 4) should be adopted [14,42,43].

The data-driven nature of ML accident analysis/classification applications necessitates the meticulous data management process (Fig. 5), precise data labeling, and curation according to the required analysis framework to ensure that sufficient and representative data is collected for the data sets that are used for training, validation and testing purposes [14].

6. Conclusions

Artificial intelligence (AI) applications are already integrated and expected to expand within the aviation sociotechnical ecosystem [14, 44–47]. EASA risk classification use cases of ML applications are anticipated as deliverables in the following concept paper [14]. Albeit preliminary, this study explored and proved that an open-source

Table 3
Comparative tables between SMEs' and AI's HFACS analysis for Helios Airways Flight HCY552.

Helios Airways Flight HCY552	
Human SMEs	ChatGPT
Organizational influences	
The airline's management did not provide adequate oversight (**SI001**). Moreover, the airline's command and control resources were deficient (**OR001**) and Cyprus DCA lacked qualified personnel (**OS002**). The operator has not provided hypoxia training (**SI003**). Also, the organizational culture within the airline was conducive to non-compliance with established safety procedures and protocols (**OC001**).	**OR001: Command and Control Resources are Deficient** The airline's command and control resources were deficient, as the airline did not have an effective safety management system in place to identify and mitigate safety risks. **OC001: Organizational Culture (attitude/actions) Allows for Unsafe Task/Mission** The organizational culture within the airline was conducive to non-compliance with established safety procedures and protocols. The airline did not place a high enough value on safety and did not provide adequate resources or training to ensure that safety protocols were followed.
Supervision	
There was inadequate risk assessment (**SP006**).	**SI001: Supervisory/Command Oversight Inadequate** The airline's management did not provide adequate oversight or support to ensure that the flight crew followed proper procedures and protocols. The airline did not have an effective safety management system in place to prevent such accidents.
Preconditions	
The First Officer did not retain procedural knowledge after training (**PC109**). The captain failed to ensure that the pre-flight checklist was completed correctly (**PP101**). The crew became confused, and task saturated with the equipment cooling fan situation (**PC103** & **PC104**). The use of the same aural warning for two different trigger events indicates warning system issues (**PE202**). The crew misinterpreted the auditory cues (**PC507**). There was lack of effective communication between the crew members in the malfunction diagnosis (**PP108**). The lack of pressurization led to hypoxia (**PC312**) and eventually loss of consciousness (**PC304**).	**PC307: Fatigue** The captain was fatigued at the time of the accident, which likely contributed to his failure to respond appropriately to the situation. **PC508: Spatial Disorientation** The flight crew likely experienced spatial disorientation due to the loss of consciousness resulting from hypoxia, which may have prevented them from recognizing the aircraft's attitude and altitude. **PP101: Failure of Crew/Team Leadership** The captain failed to ensure that the pre-flight checklist was completed correctly, and he did not recognize the pressurization system's failure or take appropriate corrective action. The captain's failure to lead his crew contributed to the accident.
Acts	
The flight crew did not complete the pre-flight checklist (**AE102**), ignored the warning indications (**AE105**), and did not follow the correct procedures when responding to the pressurization system failure (**AE103**). The crew failed to recognize the seriousness of the situation (**AE201**) and followed wrong choice of action (**AE206**).	**AE102: Checklist Not Followed Correctly** The flight crew did not complete the pre-flight checklist and failed to recognize that the pressurization system was set to manual instead of automatic mode. This error led to the aircraft cabin not pressurizing properly, which caused the crew and passengers to experience hypoxia and eventually led to the loss of consciousness of the crew. **AE103: Procedure Not Followed Correctly** The flight crew did not follow the correct procedures when responding to the pressurization system failure. They misinterpreted the warning indications, and as a result, they were unable to take the necessary corrective actions. **AE105: Ignored a Caution/Warning** The flight crew ignored the warning indications and failed to respond to the aircraft's systems and procedures indicating that the pressurization system was not functioning correctly. **AE201: Inadequate Real-Time Risk Assessment** The flight crew failed to perform an adequate real-time risk assessment, which would have identified the pressurization system failure and the resulting cabin depressurization. The crew failed to recognize the seriousness of the situation, and the delay in taking appropriate action led to the loss of the aircraft and all onboard. **AV001: Performs Work-Around Violation** The flight crew attempted to perform a work-around by manually controlling the pressurization system, which was not the correct procedure. This work-around led to the aircraft cabin not pressurizing properly, which caused the crew and passengers to experience hypoxia.

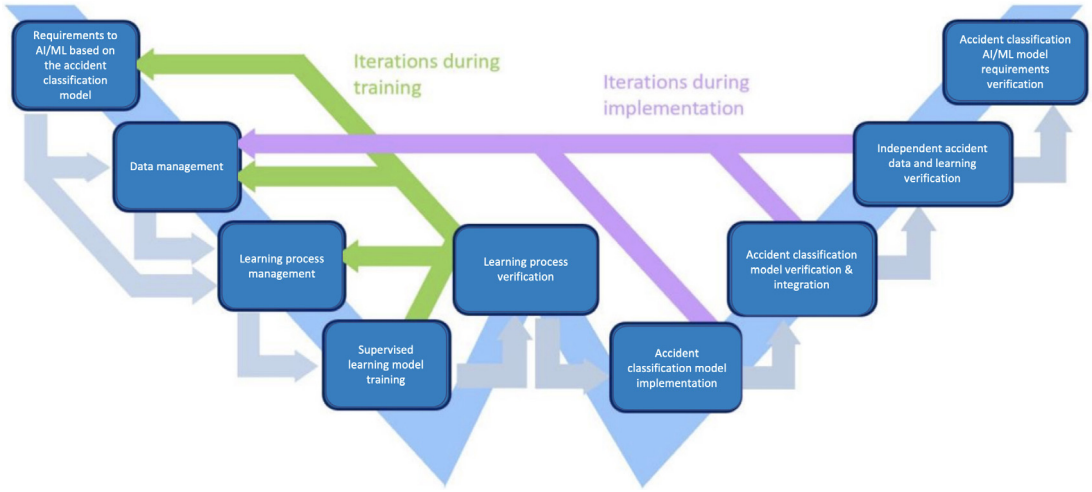

Fig. 4. Learning assurance W-shaped iterative process for AI/ML accident analysis/classification models as adapted from EASA (2023).

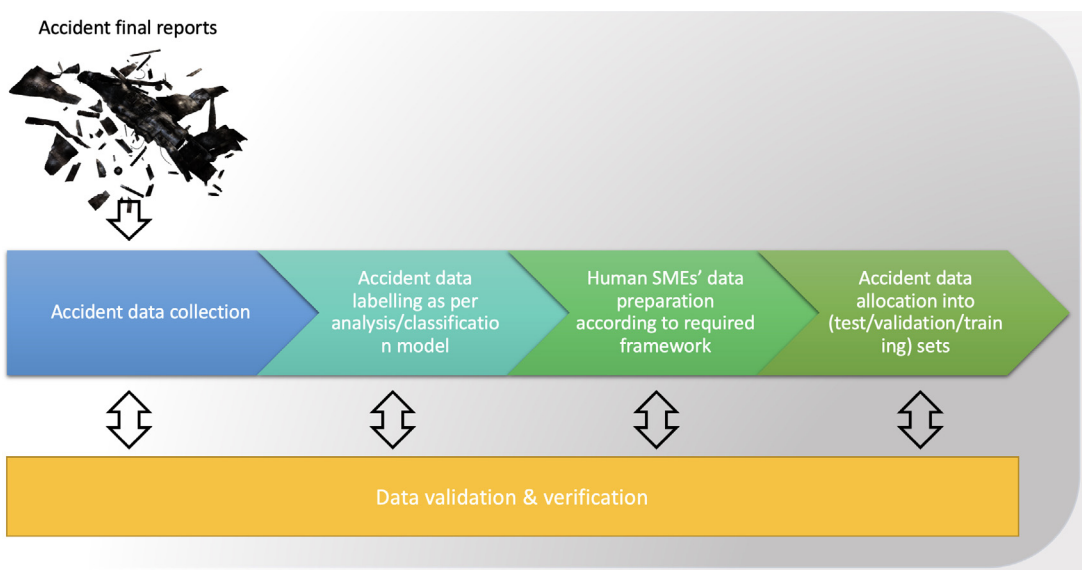

Fig. 5. Data management process for AI/ML accident analysis/classification models as adapted from EASA (2023).

AI/ML (i.e., ChatGPT) have the potential to conduct an accident analysis/classification using system-based models and is susceptible to training in the specific area. To conclude:

- AI/ML applications could facilitate the regulation for threat identification of sound safety management systems [6] and contribute to learning organizations' continuous learning, growth, and improvement by creating new knowledge from previous successes and failures utilizing appropriate methods and tools [9].
- The use of sound prompt engineering [41,48,49] and well-established classification frameworks could reach at least a "fair agreement" between human SMEs and AI in accident analysis.
- The ChatGPT performed better on linear models (i.e., Bowtie) compared to complex systemic accident causation models (e.g., HFACS).
- This version of ChatGPT could not conduct high- and low-level analysis concurrently.
- A multidisciplinary team including human factors, IT, and accident investigators experts should develop and oversee the W-shaped learning assurance process focusing mainly on data management.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT (OpenAI) in order to assess its potential implementation on aviation accident analysis and classification. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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